

Global Coherence versus Local Projection Defects

Rare B-Decay Anomaly Diagnostics in MAAT v1.6

Christof Krieg

MAAT Project – Independent Research

<https://maat-research.com>

2026

Abstract

Rare electroweak-penguin decays of beauty hadrons are among the most sensitive indirect probes of physics beyond the Standard Model. Recent LHCb analyses of $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ continue to show tensions in angular observables, especially in the P'_5 sector, while CERN and LHCb emphasize that more data and improved theory are required before any new-physics interpretation can be claimed.

This note asks a narrow methodological question: can MAAT v1.6 be used as a structural diagnostic layer that separates coherent global anomaly patterns from localized hadronic projection effects, without claiming new particles or replacing Standard Model effective-field-theory analyses?

We formulate a diagnostic over binned rare-decay residuals in q^2 and observable space. In this language, a new-physics-like contribution should appear as a low-cost coherent mode in Wilson-coefficient space, while a charm-loop or hadronic projection effect should appear as a more localized defect in q^2 . A synthetic toy benchmark illustrates the distinction: the coherent Wilson-shift pattern has high globality (0.9161) and low locality (0.1586), whereas the localized charm-like projection has low globality (0.0139) and high locality (0.6894). A q^2 -bin shuffled null test preserves residual amplitudes while disrupting bin coherence. These numbers are not LHCb measurements. They only demonstrate a reproducible diagnostic protocol.

The proposal is conservative: MAAT v1.6 can act as a pre-fit anomaly triage layer, ranking whether residual structure is globally coherent across bins and observables or concentrated in local projection channels. This is a synthetic proof-of-concept; a real test requires published binned LHCb or flavour-observable data and covariance matrices. It provides a reproducible structural ranking, not a physical interpretation or truth score.

1 Status and Guardrail

This is an extra phenomenological note with three guardrails:

- It does not claim that MAAT explains the current B-anomaly tensions.
- The benchmark is synthetic and uses no numerical LHCb, CMS, Belle, Belle II, or HEPData measurements.
- A real analysis would require the published binned observables, covariance matrices, form-factor uncertainties, and a professional effective-Hamiltonian fit.

The scientific question is not

Does MAAT prove new physics?

but rather

Can structural diagnostics help classify anomaly patterns before over-interpreting them?

2 Rare B Decays as Projection-Sensitive Channels

Flavour-changing neutral-current processes such as $b \rightarrow s\ell^+\ell^-$ are loop-suppressed in the Standard Model. They are therefore sensitive to small additional amplitudes, but they are also theoretically demanding because hadronic effects can mimic or obscure short-distance signals.

The decay

$$B^0 \rightarrow K^{*0}\mu^+\mu^-$$

is especially useful because its angular distribution provides multiple observables across dilepton-invariant-mass bins q^2 . LHCb has reported a persistent tension in the angular observable P'_5 , below the five-sigma discovery standard, and stresses that improved theory and Run-3 data are needed to clarify the origin of the pattern [1]. Nature’s 2026 news explainer describes the tension as being around the four-sigma level and discusses both possible new-particle interpretations and the caution required because of “charming-penguin” hadronic effects [3].

This is exactly the type of situation for which a structural diagnostic can be useful. The issue is not simply whether there is a residual. The issue is what kind of residual geometry it has:

- Is the residual coherent across many q^2 bins and observables?
- Does it align with a low-dimensional Wilson-coefficient deformation?
- Is it localized near hadronic regions or resonance-sensitive bins?
- Does it persist under channel splits, bin removals, and shuffled-null tests?

In this note, a “projection defect” means a long-distance hadronic residual projected onto a small set of measured angular observables and q^2 bins. The intended examples are charm-loop and “charming-penguin” contributions that can imitate part of a short-distance anomaly pattern. The term is not introduced as a new particle or as a new fundamental field.

3 Effective Hamiltonian Language

The standard short-distance description is written in terms of an effective Hamiltonian,

$$\mathcal{H}_{\text{eff}} = -\frac{4G_F}{\sqrt{2}}V_{tb}V_{ts}^* \sum_i C_i(\mu) \mathcal{O}_i(\mu), \quad (1)$$

where the Wilson coefficients C_i encode short-distance physics and the operators $\mathcal{O}_i$ encode the allowed low-energy interactions. For $b \rightarrow s\ell^+\ell^-$ transitions, the coefficients C_9 and C_{10} are often central in phenomenological interpretations.

A coherent new-physics-like pattern would typically be expressible as a small displacement

$$C_i \rightarrow C_i + \Delta C_i \quad (2)$$

that improves many bins and observables simultaneously. A local hadronic projection effect, by contrast, may improve only a subset of bins or mimic a coefficient shift only after projection through uncertain long-distance amplitudes.

The MAAT question is therefore:

Does the observed residual look like a coherent low-dimensional deformation, or like a localized projection defect?

4 MAAT v1.6 Diagnostic Layer

MAAT v1.6 treats complex inference as search geometry. In this setting, a candidate anomaly interpretation is a point or trajectory in a space of residual patterns, nuisance modes, and effective coefficients. The diagnostic does not decide truth. It ranks structural plausibility before one spends expensive modelling effort.

Technically, in this note MAAT v1.6 is used as a bounded multi-support loss-design layer: candidate residual patterns are mapped to diagnostic supports (H, B, S, V) , combined with the emergent robustness closure R_{rob} , and ranked by a logarithmic cost. It is therefore closer to a structured pre-fit scoring functional than to a new dynamical law, Bayesian likelihood, or particle-physics model. The broader formalism is documented in [4, 5].

Let $r_{\alpha b}$ denote a normalized residual or pull for observable α in bin b . The residual matrix is

$$R = \{r_{\alpha b}\}.$$

We define four operational supports:

Support	Meaning here	Operational role
H	Effective-Hamiltonian coherence	Does a low-dimensional global coefficient mode explain the residual?
B	Bin-balance and smoothness	Does the pattern vary smoothly in q^2 rather than forming isolated spikes?
S_{eff}	Bounded anomaly activity	Is the residual strong enough to be informative without becoming uncontrolled?
V	Shared residual coherence	Do observables share a low-dimensional deformation mode rather than independent bin noise?

The v1.2.1 robustness closure is retained:

$$R_{\text{resp}} = (HBV)^{1/3}, \quad R_{\text{rob}} = \min \left[R_{\text{resp}}, (HBS_{\text{eff}}V)^{1/4} \right]. \quad (3)$$

The diagnostic cost is

$$F_{\text{diag}} = -\log(\epsilon + H) - \log(\epsilon + B) - \log(\epsilon + S_{\text{eff}}) - \log(\epsilon + V) - \log(\epsilon + R_{\text{rob}}). \quad (4)$$

Lower F_{diag} indicates lower diagnostic stress under the proposed score. It does not imply correctness.

5 Globality and Locality

Two companion quantities make the interpretation explicit.

First, define a low-dimensional global-mode fit R_{global} . In the toy implementation this is only a constant q^2 -independent mode per observable:

$$R_{\text{global}, \alpha b} = \bar{r}_{\alpha}. \quad (5)$$

Then the globality score is

$$G_{\text{mode}} = 1 - \frac{\|R - R_{\text{global}}\|_2}{\|R\|_2 + \epsilon}. \quad (6)$$

Second, define a locality pressure from the upper tail of the residual after removing the global mode:

$$L_{\text{tail}} = \frac{q_{95}(|R - R_{\text{global}}|)}{q_{95}(|R - R_{\text{global}}|) + \|R\|_2 + \epsilon}. \quad (7)$$

A new-physics-like pattern should tend toward high G_{mode} and lower L_{tail} . A local projection or hadronic-nuisance pattern should tend toward lower G_{mode} and higher L_{tail} . This is the central classification idea. Different physical mechanisms may still project onto similar diagnostic coordinates; therefore the score can prioritize modelling effort, but it cannot uniquely identify the microscopic origin of an anomaly.

6 Synthetic Toy Benchmark

The script constructs four synthetic residual patterns over representative q^2 bins and three abstract observables:

- Standard-Model-like noise.
- A coherent Wilson-coefficient-like shift.
- A localized charm-like projection distortion.
- A mixed coherent-plus-local pattern.

No external experimental data are used. The purpose is only to show that the proposed coordinates behave in the intended direction.

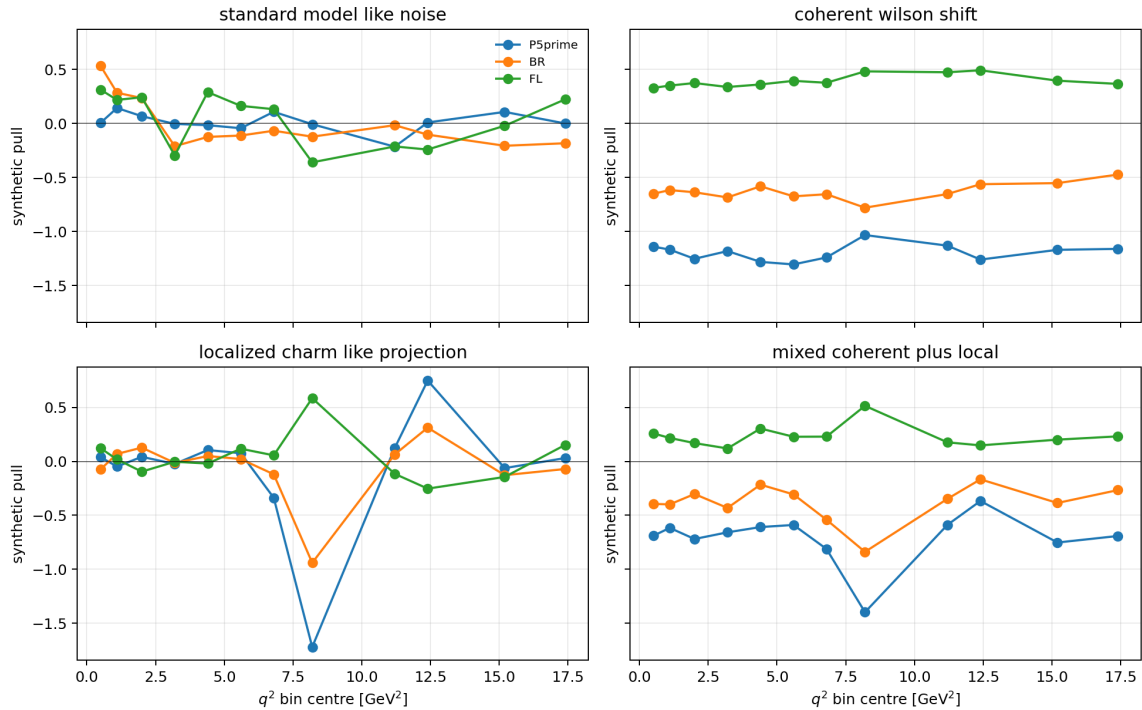


Figure 1: Synthetic residual patterns. The coherent Wilson-shift pattern is broad across q^2 , while the charm-like projection pattern is localized.

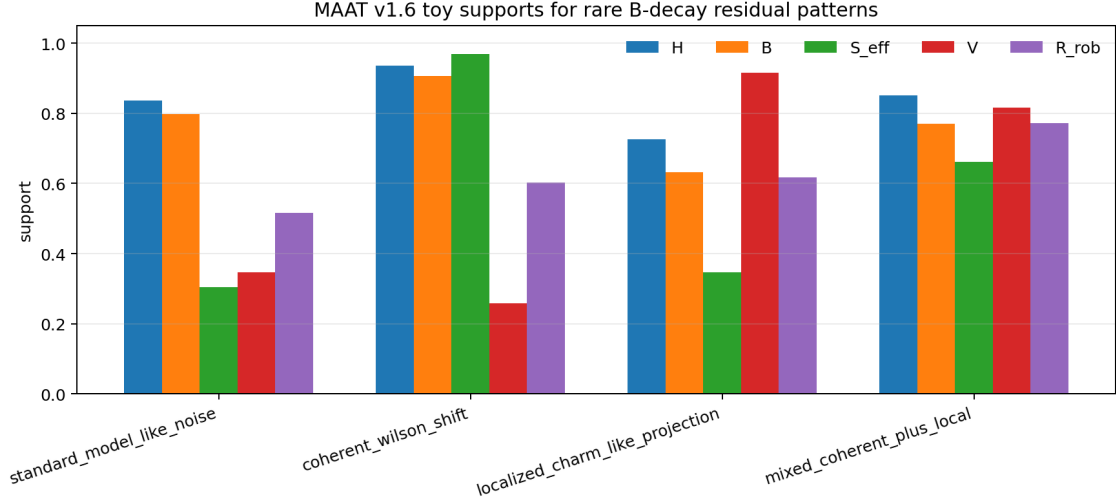


Figure 2: MAAT supports for the synthetic residual patterns. The supports are diagnostic coordinates only; they are not fitted particle-physics parameters.

Scenario	H	B	S_{eff}	V	G_{mode}	L_{tail}
SM-like noise	0.8362	0.7980	0.3051	0.3471	0.0074	0.6394
Coherent Wilson shift	0.9361	0.9069	0.9695	0.2573	0.9161	0.1586
Localized charm-like projection	0.7269	0.6323	0.3468	0.9168	0.0139	0.6894
Mixed coherent plus local	0.8510	0.7704	0.6614	0.8160	0.6594	0.4178

Table 1: Synthetic diagnostic summary. The table illustrates the intended separation between coherent global deformation and localized projection pressure.

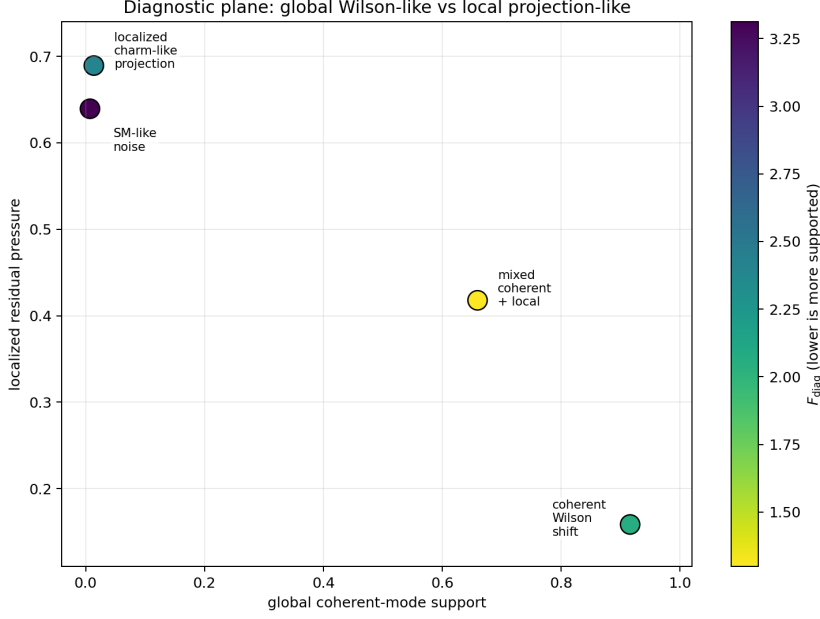


Figure 3: Diagnostic plane for the central distinction of the note: global coherence versus local projection pressure. The horizontal direction tracks support for a coherent coefficient-like mode, while the vertical direction tracks localized residual pressure after the global mode is removed. A real analysis would replace these toy patterns with published binned residuals and covariance matrices.

6.1 Bin-Shuffled Null Test

To test whether the diagnostic responds to residual geometry rather than only to residual amplitude, the script also performs a q^2 -bin shuffled null test. For each synthetic scenario, the bin order is independently permuted within each observable. This preserves the set of residual values while disrupting coherent bin-to-bin structure.

Scenario	Real F_{diag}	Null mean F_{diag}	Null mean ΔF_{diag}
SM-like noise	3.3125	4.0090	0.6965
Coherent Wilson shift	2.0594	2.2417	0.1823
Localized charm-like projection	2.4041	4.4619	2.0578
Mixed coherent plus local	1.2985	3.1323	1.8338

Table 2: Bin-shuffled null summary with 200 shuffles per scenario. The null test is intentionally simple: it preserves amplitude distributions but destroys coherent bin ordering. In this toy setup the simple globality and locality summaries are amplitude-invariant under bin shuffling, while the full diagnostic cost increases through the smoothness and connectivity channels.

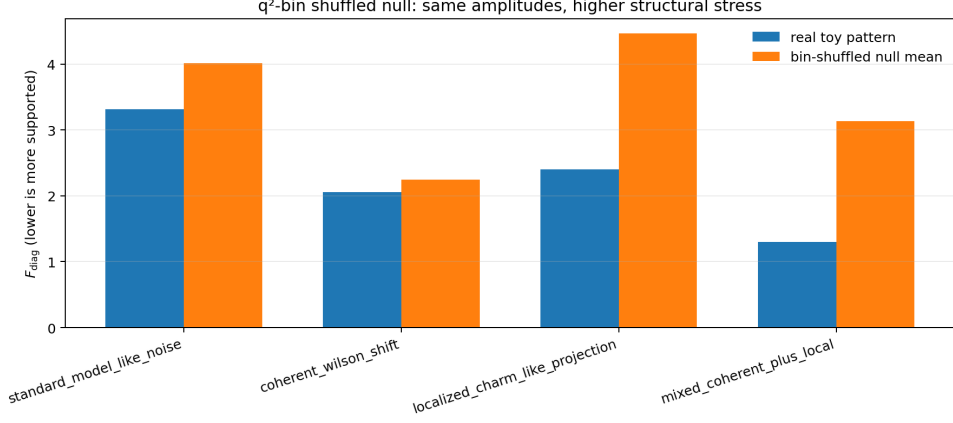


Figure 4: q^2 -bin shuffled null comparison. The null preserves residual amplitudes but generally raises F_{diag} , showing that the score is sensitive to bin coherence through the B and V channels. Future real-data tests should use richer Wilson-mode templates and channel-holdout nulls.

7 How a Real Test Would Work

A genuine analysis would require:

- published binned observables such as P'_5 , F_L , branching fractions, and lepton-universality ratios where available;
- experimental covariance matrices and bin correlations;
- form-factor and charm-loop nuisance models;
- competing Standard Model, nuisance-only, and Wilson-coefficient-shift fits;
- shuffled-bin, shuffled-observable, and channel-holdout null tests.

Operationally, one would build a residual vector

$$r_i = (O_i^{\text{exp}} - O_i^{\text{SM}}),$$

with covariance matrix Σ_{ij} over all selected bins and observables. The residual norm and all global-mode fits should then be evaluated using the whitened inner product

$$\|r\|_{\Sigma}^2 = r_i (\Sigma^{-1})_{ij} r_j.$$

The MAAT coordinates would be computed on the whitened residual vector or on its best-fit Wilson-coefficient and nuisance-model projections. A real version of G_{mode} would compare the covariance-weighted residual before and after fitting a low-dimensional coefficient displacement such as $(\Delta C_9, \Delta C_{10})$, while L_{tail} would test whether the remaining whitened residual is concentrated in a small number of q^2 bins or channels.

The next non-toy version would therefore be a binned-data pilot with three predeclared inputs: (i) an experimental residual vector for selected $B \rightarrow K^{(*)}\ell^+\ell^-$ observables, (ii) the corresponding covariance matrix, and (iii) two baseline projection families, one Wilson-like and one long-distance nuisance-like. The MAAT contribution would then be judged only by whether it improves predeclared coherence and shuffled-null tests beyond those baselines.

The comparison target is not a full flavour-fit package such as `flavio`, `EOS`, `SuperIso`, `HEPfit`, or `FlavBit`. Those tools remain the correct instruments for parameter inference. The MAAT diagnostic would sit upstream of them as a pre-fit residual-geometry screen: it ranks whether a

residual pattern is more consistent with a coherent Wilson-like deformation, a localized hadronic projection, or a mixed structure. Its value would be measured only by whether this pre-fit ranking agrees with, accelerates, or usefully stress-tests the subsequent full likelihood analysis.

The falsifiable test would be:

Does a low-dimensional structural mode improve anomaly coherence across bins and observables more than nuisance-only, local-bump, and shuffled-null models?

If yes, MAAT would not prove new physics, but it would identify a coherent global residual worthy of further fit effort. If no, the anomaly would be structurally consistent with local projection effects or current theory uncertainty.

8 Interpretation

The useful distinction is not “Standard Model versus MAAT”. The useful distinction is:

global coefficient-space defect versus local projection defect.

This distinction is already part of the flavour-physics discussion in more standard language. MAAT v1.6 adds a structural ranking layer:

- coherent global patterns receive higher H and lower locality pressure;
- local hadronic distortions receive lower globality and higher tail pressure;
- cross-observable persistence is measured separately through V ;
- robustness is emergent rather than imposed as a fifth primary sector.

The important point is negative as much as positive. A residual is not automatically new physics. A residual becomes more interesting when its geometry is coherent, low-dimensional, stable under ablations, and resistant to shuffled-null explanations.

9 Limitations

- The numerical benchmark is synthetic and contains no LHCb data.
- The global mode is simplified to a constant q^2 pattern rather than a professional Wilson-coefficient fit.
- No covariance matrices, form-factor uncertainties, or charm-loop models are included.
- Different physical mechanisms may project onto similar values of G_{mode} and L_{tail} ; these coordinates are not unique microscopic identifiers.
- The support definitions are heuristic diagnostics, not derived from QCD or an effective field theory, and should therefore be understood as a complementary diagnostic layer rather than a replacement for standard EFT global fits.
- The framework can rank anomaly patterns but cannot decide whether a new particle exists.

10 Reproducibility

The toy benchmark is reproducible with:

```
python3 rare_b_decay_anomaly_diagnostics.py
```

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Related MAAT background notes in the same repository include *Core Formalism of Structural Selection* and *MAAT v1.6: Structural Search Geometry*.

Suggested folder:

```
experiments/rare_b_decay_anomaly_diagnostics/
```

Generated outputs include:

- `toy_diagnostic_summary.csv`
- `toy_diagnostic_summary.json`
- `toy_residual_patterns.csv`
- `toy_bin_shuffled_null_samples.csv`
- `toy_bin_shuffled_null_summary.csv`
- three diagnostic figures.
- one bin-shuffled null figure.

For reproducibility, the main script and compact summary artifacts have the following SHA256 prefixes; full SHA256 hashes are listed in the experiment README:

Artifact	SHA256 prefix
<code>rare_b_decay_anomaly_diagnostics.py</code>	ae8cf785873f0e25
<code>toy_diagnostic_summary.csv</code>	3d72aa8b8ac83c9e
<code>toy_diagnostic_summary.json</code>	9cc69182ea710f20
<code>toy_bin_shuffled_null_summary.csv</code>	6ebd366e9d7e4854

11 Data Attribution and License Notes

No LHCb, CMS, Belle, Belle II, or HEPData numerical data are redistributed in this repository for the toy benchmark. No CERN, LHCb, Nature, Physical Review Letters, or HEPData figures, tables, images, or long text passages are reused. The paper cites public scientific sources for context only and paraphrases the state of the field. Any future analysis using experimental tables, plots, covariance matrices, figures, images, or HEPData records must cite the relevant collaborations, papers, records, and licenses directly.

No endorsement by CERN, LHCb, CMS, Belle, Belle II, HEPData, Physical Review Letters, Nature, or the cited authors is implied.

12 Conclusion

Rare B-decay anomalies are a natural testbed for structural diagnostics because they sit exactly at the boundary between possible short-distance new physics and difficult long-distance projection effects.

This note proposes a limited MAAT v1.6 role: not to explain the anomaly, but to classify residual geometry. In the toy benchmark, coherent Wilson-like shifts and localized charm-like projection patterns occupy different regions of the diagnostic plane. A real test would require published binned LHCb or flavour-observable data, covariance matrices, and comparison against professional flavour-fit baselines. The present result is therefore only a proof-of-concept with synthetic residual patterns.

The practical contribution is therefore a guardrailed protocol:

rank anomaly structure before claiming anomaly meaning.

References

- [1] LHCb Collaboration, “Searching for new physics with beauty particles,” CERN News, 24 September 2025. <https://home.cern/searching-new-physics-beauty-particles/>.
- [2] R. Aaij et al. (LHCb Collaboration), “Comprehensive analysis of the $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ decay,” publicly discussed in CERN and Nature context summaries (2026); exact journal metadata should be verified against the final collaboration record before citation.
- [3] E. Gibney, “The exotic particles that could finally break the standard model,” *Nature News* Explainer, 1 May 2026; correction 19 May 2026. DOI: [10.1038/d41586-026-01387-x](https://doi.org/10.1038/d41586-026-01387-x).
- [4] C. Krieg, “Core Formalism of Structural Selection,” MAAT Project documentation, preprint note (2026).
- [5] C. Krieg, “MAAT v1.6: Structural Search Geometry,” MAAT Project documentation, preprint note (2026).
- [6] G. Buchalla, A. J. Buras, and M. E. Lautenbacher, “Weak Decays Beyond Leading Logarithms,” *Reviews of Modern Physics* **68**, 1125–1144 (1996). DOI: [10.1103/RevModPhys.68.1125](https://doi.org/10.1103/RevModPhys.68.1125).
- [7] S. Descotes-Genon, J. Matias, M. Ramon, and J. Virto, “Implications from clean observables for the binned analysis of $B \rightarrow K^* \mu^+ \mu^-$ at large recoil,” *Journal of High Energy Physics* **2013**, 048 (2013). DOI: [10.1007/JHEP01\(2013\)048](https://doi.org/10.1007/JHEP01(2013)048).